SmartVista Solution

Hardware Sizing for Bancoppel

HARDWARE SIZING

VERSION 1.2

5 September 2023

bpcbt.com

Notice

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Contents

THE LIST OF AUTHORS AND DATES OF UPDATE

Surname Name	Position	Ver	Comment	Date	Internal Number
Moskalenko Alexey	System Architect	1.0	Initial document or generated document	01.06.23	BCS-10027
Andrei Volokhonskyi	System Architect	1.1	* Changed OS version * Changed Weblogic version * CG moved to CLOUD	25.08.2023	BCS-10196
Andrei Volokhonskyi	System Architect	1.2	* Added software in section 3.2.1, 3.2.2	05.09.2023	BCS-10256

1 Introduction

The following elements of the SmartVista solution are proposed:

- **¡Error! No se encuentra el origen de la referencia.**
- 3 Specification

> **Note:** The hardware specification will be finalized after the FSD is completed.

2 Business requirements

Category	Name	Value
General	Architecture diagram	Yes
	Propose hardware	Yes
	Oracle license quantity calculation	Yes
	Who will delivery Oracle licenses	Customer
	SmartVista Components	Front End • Front-End App • Front-End DB Back Office • Back Office DB • SVBUS App GUI • GUI App Card Generator • CG App • CG DB SV Integration Platform • SVIP App • SVIP DB • Elasticsearch • Logstash • Kibana • Keycloak
	TPS – Peak	1
	TPS – Average	0.5
	TPD – Average	6 500
	Cards	450 000
	Accounts (Clients)	150 000
	Merchants	N/A
	Total number of POSs	N/A
	Total number of ATMs	N/A
	Visa	Yes
	MasterCard	No
	China Union Pay	No
	Dinners Club	No
IT Infrastructure	PCI-DSS	Yes
	DR Site	Yes
	CG on DR Site	No
	Number of non-Prod environments	3
	Test (SIT) environment size in % of PROD	10
	Development (DEV) environment size in % of	10
	Pre-Prod (UAT) environment size in % of	50
	non-Prod environments location	Main Site
	Clustering on Main Site	None
	CG Clustering	None
	Clustering on DR Site	None
	Pre-Prod Clustering	None

Category	Name	Value
	Backup to Tapes Include	No
	Zabbix Monitoring	CloudWatch + Graphana (for transactional data)
	Proxy server	No
	Authorization service	No
	Hardware Platform App	CLOUD
	Hardware Platform DB	CLOUD
	Virtualization platform App	CLOUD
	Virtualization platform DB	CLOUD
FE Front End	Only the switch functionality	No
	Database type	Oracle
	Retention period for **fe-app** logs plain (days)	1
	Retention period for **fe-app** logs gzipped	7
	Retention period for **fe-db** data (days)	180
	NonStop - Connection to external systems	0
BO Back Office	EOD (End Of Day) duration (hours)	12
	Only the clearing functionality (Yes/No)	No
	Credits (Yes/No)	Yes
	Retention period for **bo-db** data (days)	180
	Retention period for **bo-db**log (days)	180
UI User Interface	**Total** number of sv-ui users	20
	Total number of **concurrent** users BO part	5
	Total number of **concurrent** users FE part	5
	Retention period for sv-ui logs plain (days)	1
	Retention period for sv-ui logs gzip (days)	180
CG Card Generator	Cards issued per day	500
	EMV Cards	Yes
	PCI Card Production	No
	Database type	Oracle
	Application server type	Oracle WebLogic
	Retention period for cg-db data (days)	180
	Retention period for cg-db log (days)	180
SVIP SmartVista Integration Platform	TPS – Peak	1
	TPD – Average	37 500
	Database type	Oracle
	Retention period for **app** logs plain (days)	7
	Retention period for **app** logs gzip (days)	14
	Retention period for **db data** (days)	180
	Authorization service	Keycloak

3 Specification

The following elements are related to the specification:

- Software
- Minimum hardware requirements
- System architecture
- Architecture overview
- Hardware equipment

6.1 Software

Component	Version
Virtualization	Amazon AWS EC2
OS	RHEL 8.6
Cluster	Amazon Could embedded HA function (on level Cloud)
Application Server	WebLogic 14c Standard Edition 14.1.1
Java for WLS	Oracle JDK 8
Java for SVIP, ELK	Eclipse TemurinJDK 17
Java for Keycloak	OpenJDK 11
SVIP Logging	ELK (Elasticsearch, Logstash, Kibana)7.17
SVIP IAM	Keycloak 18.0.2
SVIP WebServer	Nginx 1.23.3
Database	Oracle 19c Enterprise Edition (Multitenant)
MQ software	Apache ActiveMQ 5.16.3
Monitoring solution	Grafana

6.2 Minimum hardware requirements

The hardware requirements are listed for two sites:

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Env	Component	Software	CPU type/Shape	Cores	vCPU	RAM Gb	OS+DATA, GB	FRA \ WAL, GB	TOTAL HDD, GB	HDD IOPS
Production	FE DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	BO DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	CG DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	100
	SVIP/KCK DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	FE App	Oracle Client	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	206	0	206	100
	GUI/BUSApp	Oracle JDK, Oracle Weblogic, Apache ActiveMQ	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	276	0	276	100
	CG App	Oracle JDK, Oracle Weblogic	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	176	0	176	100
	SVIP/KCKApp	Eclipse TemurinJDK, OpenJDK, Logstash, Keycloak, nginx	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	276	0	276	100
	SVIP ELK	Eclipse TemurinJDK, Elasticsearch, Kibana	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	376	0	376	100
Pre-Production	FE DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	336	300	536	60
	BO DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	336	300	536	60
	CG DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	336	300	536	60
	SVIP/KCK DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	336	300	536	60
	FE App	Oracle Client	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	206	0	206	40
	GUI/BUSApp	Oracle JDK, Oracle Weblogic, Apache ActiveMQ	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	176	0	176	40
	CG App	Oracle JDK, Oracle Weblogic	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	176	0	176	40
	SVIP/KCK App	Eclipse TemurinJDK, OpenJDK, Logstash, Keycloak, nginx	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	176	0	176	40
	SVIP ELK	Eclipse TemurinJDK, Elasticsearch, Kibana	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	276	0	276	40
Test	Test DBs	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	536	0	536	60
	Test APPs	All APPs	Intel Xeon Platinum 8375C @2.9GHz / r6i.xlarge	2	4	32	406	0	406	40

Dev	DevDBs	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	536	0	536	60
	DevAPPs	All APPs	Intel Xeon Platinum 8375C @2.9GHz / r6i.xlarge	2	4	32	406	0	406	40
'	Grafana		Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	500	0	500	100
	Bastionhost		Intel Xeon Platinum 8488C @2.4GHz / m7.xlarge	1	2	8	100	0	100	10
	Summary			***38***	***76***	***336***	***7892***	***3000***	***10492***	***2250***

6.2.1 DR site

The hardware requirements are listed for the DR site is listed in the table below.

Env	Component	Software	CPU type/Shape	Cores	vCPU	RAM Gb	OS+ DATA , GB	FRA \ WAL, GB	TOTAL HDD, GB	HDD IOPS
Production	FE DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	BO DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	SVIP/KCK DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	FE App	Oracle Client	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	206	0	206	100
	GUI/BUSApp	Oracle JDK, Oracle Weblogic, Apache ActiveMQ	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	276	0	276	100
	SVIP/KCKApp	Eclipse TemurinJDK, OpenJDK, Logstash, Keycloak, nginx	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	276	0	276	100
	SVIP ELK	Eclipse TemurinJDK, Elasticsearch, Kibana	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	376	0	376	100
'	Grafana		Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	500	0	500	100
	Bastionhost		Intel Xeon Platinum 8488C @2.4GHz / m7.xlarge	1	2	8	100	0	100	10
	Summary			***17***	***34***	***136***	***3042***	***1350***	***4392***	***1410***

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The follow legend is used:

Standalone Application (Non-Oracle WebLogic SE)	Test Application (Oracle WebLogic SE)
Oracle WebLogic SE Application	Test Database (Oracle Database EE)
Oracle Database EE	

6.2.2 Main (DC) site

The hardware requirements are listed for the main (DC) site is listed in the table below.

Env	Component	Software	CPU type/Shape	Cores	vCPU	RAM Gb	OS+DATA, GB	FRA \ WAL, GB	TOTAL HDD, GB	HDD IOPS
Production	FE DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	BO DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	CG DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	100
	SVIP/KCK DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	FE App	Oracle Client	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	206	0	206	100
	GUI/BUSApp	Oracle JDK, Oracle Weblogic, Apache ActiveMQ	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	276	0	276	100
	CG App	Oracle JDK, Oracle Weblogic	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	176	0	176	100
	SVIP/KCKApp	Eclipse TemurinJDK, OpenJDK, Logstash, Keycloak, nginx	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	276	0	276	100
	SVIP ELK	Eclipse TemurinJDK, Elasticsearch, Kibana	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	376	0	376	100
Pre-Production	FE DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	336	300	536	60
	BO DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	336	300	536	60
	CG DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	336	300	536	60
	SVIP/KCK DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	336	300	536	60
	FE App	Oracle Client	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	206	0	206	40
	GUI/BUSApp	Oracle JDK, Oracle Weblogic, Apache ActiveMQ	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	176	0	176	40
	CG App	Oracle JDK, Oracle Weblogic	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	176	0	176	40
	SVIP/KCK App	Eclipse TemurinJDK, OpenJDK, Logstash, Keycloak, nginx	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	176	0	176	40
	SVIP ELK	Eclipse TemurinJDK, Elasticsearch,	Intel Xeon Platinum 8488C @2.4GHz / m7i.large	1	2	8	276	0	276	40

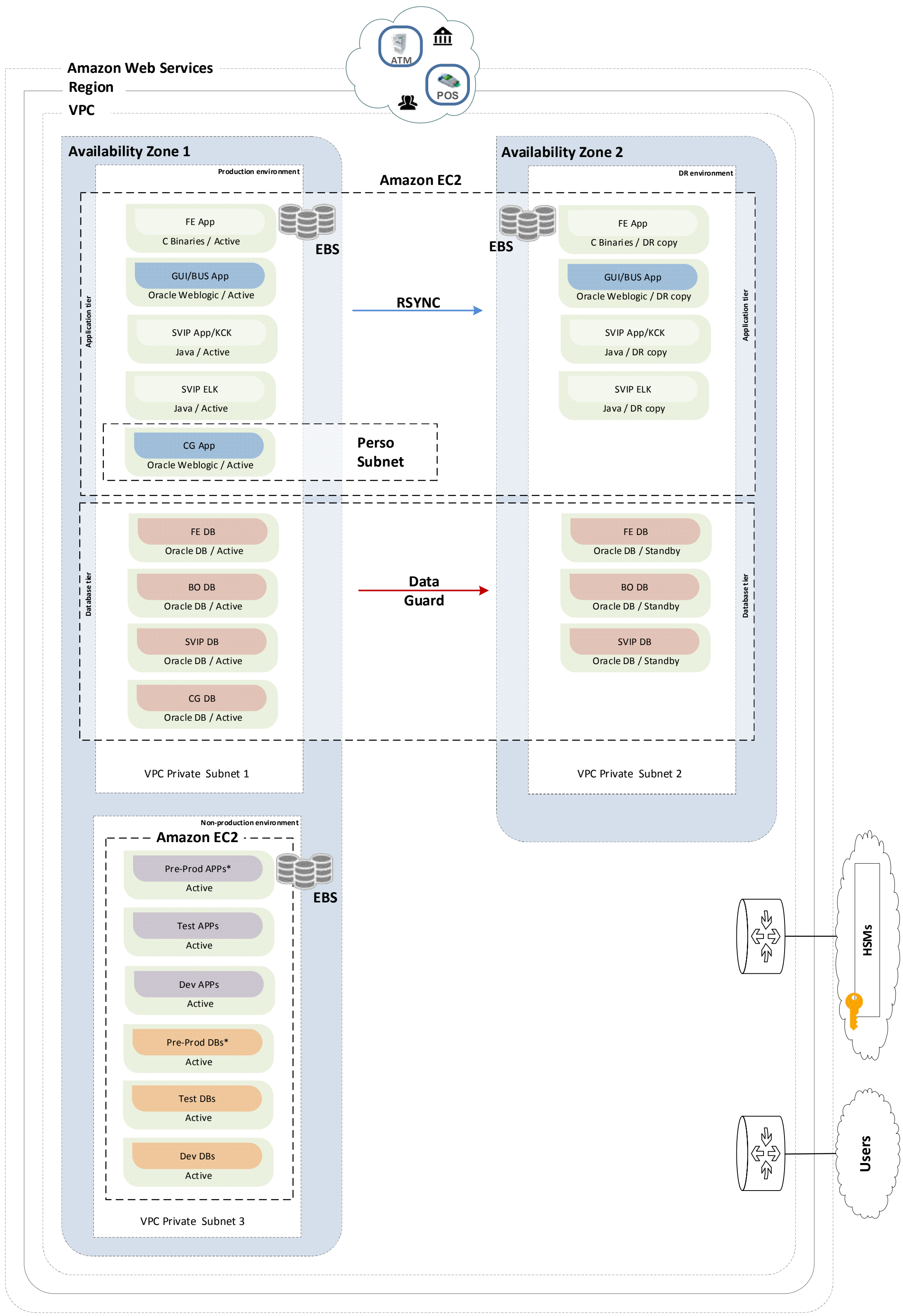
		Kibana								
Test	Test DBs	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	536	0	536	60
	Test APPs	All APPs	Intel Xeon Platinum 8375C @2.9GHz / r6i.xlarge	2	4	32	406	0	406	40
Dev	DevDBs	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	536	0	536	60
	DevAPPs	All APPs	Intel Xeon Platinum 8375C @2.9GHz / r6i.xlarge	2	4	32	406	0	406	40
-	Grafana		Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	500	0	500	100
	Bastionhost		Intel Xeon Platinum 8488C @2.4GHz / m7.xlarge	1	2	8	100	0	100	10
	Summary			***38***	***76***	***336***	***7892***	***3000***	***10492***	***2250***

6.2.3 DR site

The hardware requirements are listed for the DR site is listed in the table below.

Env	Component	Software	CPU type/Shape	Cores	vCPU	RAM Gb	OS+DATA, GB	FRA\WAL, GB	TOTAL HDD, GB	HDD IOPS
Production	FE DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	BO DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	SVIP/KCK DB	Oracle DB	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	436	450	886	300
	FE App	Oracle Client	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	206	0	206	100
	GUI/BUSApp	Oracle JDK, Oracle Weblogic, Apache ActiveMQ	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	276	0	276	100
	SVIP/KCKApp	Eclipse TemurinJDK, OpenJDK, Logstash, Keycloak, nginx	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	276	0	276	100
	SVIP ELK	Eclipse TemurinJDK, Elasticsearch, Kibana	Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	376	0	376	100
-	Grafana		Intel Xeon Platinum 8488C @2.4GHz / m7i.xlarge	2	4	16	500	0	500	100
	Bastionhost		Intel Xeon Platinum 8488C @2.4GHz / m7.xlarge	1	2	8	100	0	100	10
	Summary			***17***	***34***	***136***	***3042***	***1350***	***4392***	***1410***

6.3 System architecture



*Pre-Production environment has more than one instance. This one instance reflected just to simplify the diagram. Please refer on a sizing calculation table for more details.

6.4 Architecture overview

During the implementation of this project, the architecture will be clarified and may change. This is a high-level architecture, which should be clarified with the customer and consider all specific requirements such as payment network configuration, business requirements etc. Further discussion on this architecture between the customer representative and BPC is required and should take a place before the sizing document accepted.

This sizing document is designed to utilize x86 cloud architecture represented by Amazon AWS Cloud (AWS).

Virtualization by Amazon EC2 is used for all SmartVista applications and databases.

The Customer assumes all risks associated with AWS Cloud. The client installs, configures, monitors, and provides ready-made VMs for the BPC team. The BPC team does not configure the AWS Cloud system and "HA" based on the AWS Cloud system. If Customer has any questions related to the AWS Cloud, the Customer will contact AWS support.

Also, BPC team don’t answer on all questions about infrastructure of AWS include details about hardware in CLOUD, software in Cloud and other infrastructure details of AWS, networks, firewalls setting and other security things, because Customer must ask that information in AWS support and work with support for create TEST, DEV, Production and other environments. Infrastructure in AWS it is full responsibility of Customer.

The customer must choose "disk type" according to the expected load on virtual machines according to the calculated sizing. If the customer has questions about choosing a "disk type" and which one is more suitable according to the sizing, the Customer undertakes to contact AWS cloud support. Since everything is constantly changing in the AWS cloud, the BPC team cannot provide 100% up-to-date information.

Please contact AWS support for consultation and selection of the most optimal value of any parameters of virtual machines in the cloud according to the requirements of the calculated sizing.

All hard disks of all environments Smartvista is Amazon EBS volumes. It is block devices.

Clustering is not applied in this solution on this project.

All GUI applications are consolidated in one virtual machine,GUI App:

- Back-office GUI
- Front End GUI
- Camel converters

GUI App virtual machine resources are allocated using JVM startup parameters for each managed server in Oracle WebLogic server.

The SV-BUS application is an Apache ActiveMQ located on an application server with a graphical interface (GUI App). Java Oracle JDK is used for this configuration.

Keycloak will be install on SVIP App server.For database of Keycloak will use schema in SVIP database.

Multitenant technology is used for all Oracle databases, a PDB database is created for each database instance for each application. Attention, only 3 PDB databases can be created per database instance, if more are required, multitenant licenses must be purchased.

For test and dev environments, one PDB database is used for all applications. Each application uses its own schema or schemas. This is required to save resources.

The following Oracle options required:

Component	Database Options					
	Advanced security	Partitioning	Diagnostic Pack	Multitenant	RAC	GG
FE DB	✓	✓	✓			
BO DB	✓	✓	✓			
SVIP/KCK DB	✓		✓			
CG DB	✓		✓			

Test, Development and Pre-production environments should be installed on additional separate instances using EC2 in the Amazon AWS cloud. Pre-production environments should have exactly the same architecture as production environments, but with fewer resources. This is necessary to save resources.

AWS environment should have a connection to HSM with enough performance model using AWS VPN connect or AWSDirect Connect with minimum possible latency. HSMs are provided by on premise hardware equipment.

The second availability zone (DRsite) is organized by replicating data from the first availability zone (DC site). Data Guard will be used for this purpose. RSYNC will synchronize data for applications.

The production environment should have a connection to all payment networks used in the project. The method and implementation of the connection is the full responsibility of the customer.

The open source components of the system (for example, Keycloak, OpenJDK, Grafana etc.) are configured and maintained by the customer independently, including additional add-ons and software necessary for the operation of the system.

The customer installs and implements backups and backup systems independently. This is the full responsibility of the customer.

The Customer installs, configures and supports any chosen monitoring server, monitoring agents and all monitoring setting. This is the full responsibility of the customer.

6.5 Hardware equipment for On-Prem site

Hardware equipment is listed in the table below.

Model	Component	CPUs / Cores	Memory	On-Prem site
Thales 10k PS10-CLA-S 60 CPS "Classic Package"	Authorization HSM	-	-	1
Thales 10k PS10-PRM-S 60 CPS "Premium Package"	Personalization HSM	-	-	1
Thales 10k PS10-PRM-L 25 CPS "Premium Package"	Test HSM	-	-	1

7 Oracle licenses

Since the infrastructure will be change during all project phases, Oracle license calculation results are provided for information purposes only.

Note the following:

- **Amazon AWS architecture -** In Amazon AWS two vCPUs are as equivalent to one physical core of processor. 4 vCPU are 1 socket of CPU. If necessary, the customer can change the "shapes" of Amazon EC2. But it is worth considering that the calculation of licenses will be based on vCPU. The provided "shapes" were compared with the specified real cores. In sizing document also shows the actual real cores. It is also worth clarifying and making an accurate calculation of the Oracle license with your regional manager provided by Oracle. For questions regarding "Amazon EC2 shape", the customer needs to contact Amazon AWS Cloud Support. For details information you can examine official document from Oracle: "Licensing Oracle Software in the Cloud Computing Environment". You can find document use this link: https://www.oracle.com/us/corporate/pricing/cloud-licensing-070579.pdf

The license calculation is not final, to get a complete accurate calculation at the current time, contact your Oracle representative.

7.1 Productionenvironment

7.1.1 Processor

Product Description	Metric	DC Site	DR Site	Total	Licenses
Oracle Database Enterprise Edition	Processor	6	6	12	12
Oracle Advanced Security	Processor	6	6	12	12
Diagnostic Pack	Processor	6	6	12	12
Partitioning	Processor	4	4	8	8
WebLogic Server Standard Edition	Socket	1	1	2	2

7.1.2 Named User Plus (Prod and non-Prod Card Generator)

Product Description	Metric	On-Prem Site	DR Site	Total
Oracle Database Enterprise Edition	NUP	50	-	50
Advanced Security	NUP	50	-	50
Diagnostic Pack	NUP	50	-	50
WebLogic Server Standard Edition	NUP	10	-	10

7.2 Pre-Production environment

Product Description	Metric	DC Site	DR Site	Total
Oracle Database Enterprise Edition	NUP	100	-	100
Oracle Advanced Security	NUP	100	-	100
Diagnostic Pack	NUP	100	-	100
Partitioning	NUP	50	-	50
WebLogic Server Standard Edition	NUP	20	-	20

7.3 Dev/Test environments

Product Description	Metric	DC Site	DR Site	Total
Oracle Database Enterprise Edition	NUP	100	-	100
Oracle Advanced Security	NUP	100	-	100
Diagnostic Pack	NUP	100	-	100
Partitioning	NUP	100	-	100
WebLogic Server Standard Edition	NUP	20	-	20

8 Other conditions

Note the following conditions:

- Third party software or hardware licenses may be not included in the sizing and should be quoted by local hardware or software vendor.
- There is no special requirement to use SSD hard disks, but it may be a good option for better database performance.
- To reduce the possible overload of the OLTP database (fe-db), data should be extracted or replicated to an "archive" database by Oracle GoldenGate software or another technology.
- Important data must be placed in a different LUN from the LUN with the operating system. All LUNs must be presented to both physical servers.
- Storage space provided in this Sizing is usable space. Raw disk space should be calculated by a local hardware vendor.
- The following are considered as an external service and beyond the scope of this Technical Proposal:
  - Networking solution (connecting external users or banks to SmartVista)
  - Common IT infrastructure services such as DNS, NTP, email, and management servers
  - Hardware monitoring solution
  - Backup solution

9 Glossary

fe-app

SmartVista's Front End Application. The transaction switch is responsible for online transaction processing. It is designed to perform the following functions:

- Interact with international payments networks (such as Mastercard and Visa) and other networks for processing authorization requests.
- Interact with the core banking systems (CBSs) of the financial institutions that are hosted in the SmartVista system.
- Interact with financial institution's own switches or with those owned by other financial institutions that are connected to the SmartVista system.
- Process authorization requests from ATMs, POS devices, and other systems such as Internet banking and e-commerce systems.
- Route authorization requests according to routing configuration.
- Perform stand-in authorization when the CBS is not available.
- Apply transaction filtering rules.
- Exchange cryptographic keys with terminals to ensure secure communication.
- Log and monitor transactions.
- Apply online fees and limits.

fe-db

Front End Database. It is the database that the Front End Application works with.

Bo-db

Back Office Database. SmartVista Back Office is responsible for financial processing, clearing, accounting, and reporting. SVBO performs the following high-level tasks:

- Full cycle card management, including personalization (together with specialized components).
- Credit functionality for accounts, including interest calculation, penalties, invoicing, and repayments.
- Management of customer's card, account and other related information
- Offline financial processing and clearing.
- Interaction with international and domestic payment networks, and optionally with a CBS.
- Interaction with other SmartVista modules including SVFE and SmartVista Card Generator (SVCG).
- Notifications.
- Report and statement generation.

bus

Queue software used as integration layer between SmartVista components.

Sv-ui

Java-based Graphical User Interface for all SmartVista components.

Cg-app

CardGen (Card Generation) application. It is a Java-based application used to prepare data for card issuing.

Svip

SmartVista integration platform is a packaged set of services, tools, and technologies that enable customers to integrate their different products and to extend the functionality of SmartVista solutions. The software solution processes multiple data formats, performs flexible message routing using BPM Orchestrator, and implements business logic via hosted and external service calls.

Svip-app

SmartVista Integration Platform provides data format transformation, flexible workflow routing using BPM Orchestrator, and the execution of pre-configured business logic.

Svip-db

Database that the SVIP application works with.

UAT/Pre-Prod apps

Pre-Production/UAT Server for all applications. It should be a replica of the Production environment.

Test apps

Test server for all applications.

Dev apps

Development Server for all applications.

UAT/Pre-Prod DBs

Pre-Production/UAT Server for all databases. It should be a replica of Production environment.

Test DBs

Test server for all databases.

Dev DBs

Development server for all databases.